Q2 2023

Web3 Development Report

alchemy

TL;DR

The state of web3 development

Q2 Executive Summary

- **Web3 Devs:** SDK installs and EVM smart contracts continued to climb, pointing to healthy developer activity
- **DeFi + NFTs:** Consumer DeFi and NFT trading volumes slumped, but the number of DeFi users is up 35% Q/Q
- **Three Transitions:** Security, account abstraction, and ZK-rollups gain traction following Vitalik's Three Transitions
- **Account Abstraction:** Active Account Abstraction users surge 27,360% as ERC-4337 gains traction
- **Testnets:** Sepolia will soon be the preferred developer testnet as Goerli supply dries up

Web3 dev activity continued to climb 📈

Y/Y and Q/Q comparisons highlight sustained growth in dev activity.

26.8M

Ethereum SDK installs

+37% Y/Y +7% Q/Q

11.1M

Wallet SDK installs

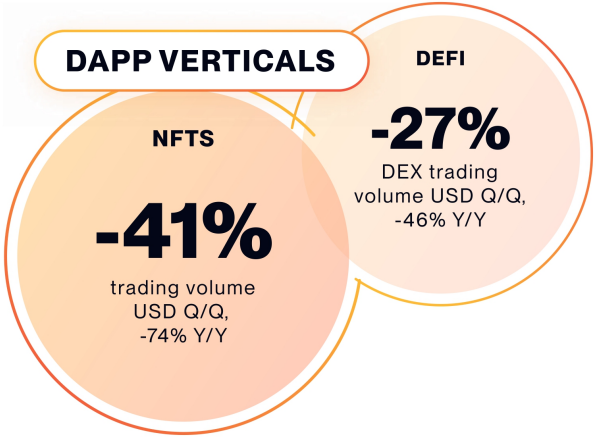
+196% Y/Y +22% Q/Q

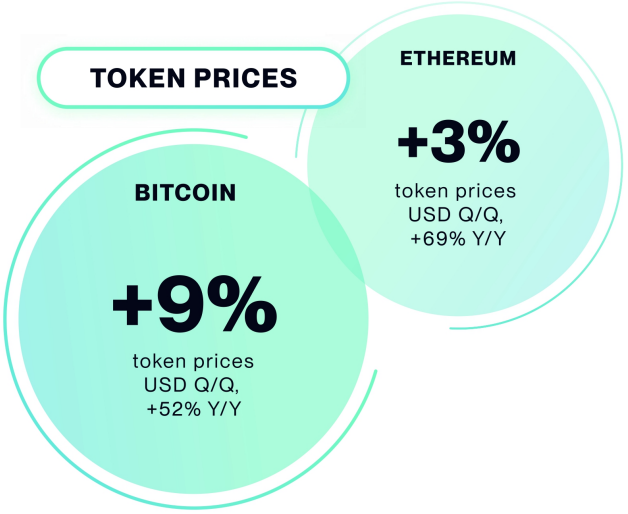
75.9M

EVM Smart Contracts created across Ethereum Mainnet, Arbitrum, Optimism and Polygon

+1,106% Y/Y +302% Q/Q

Trading volume dips, but token prices improve



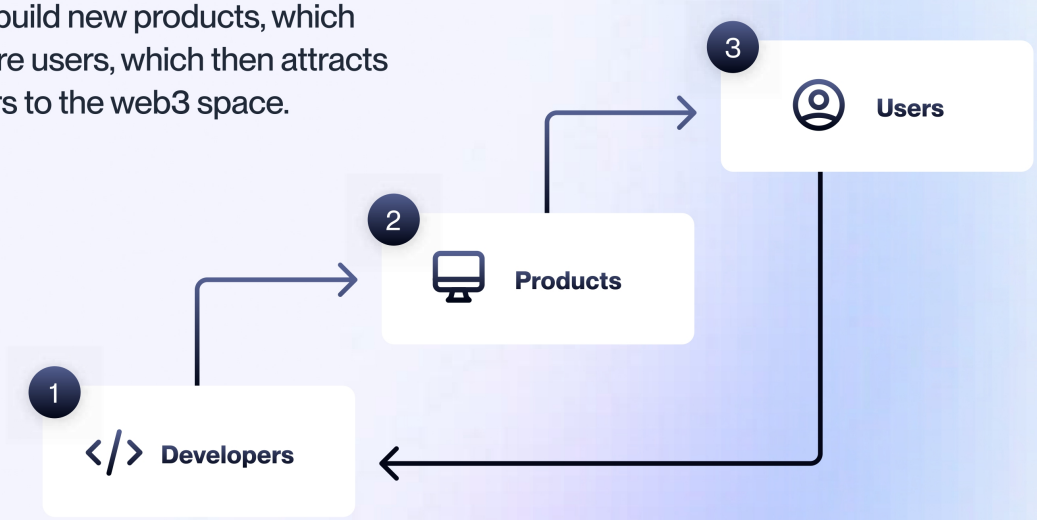


Our approach

Understanding the builder ecosystem

The development cycle

Developers build new products, which onboard more users, which then attracts more builders to the web3 space.



5 stages of developer activity

DEVELOPMENT STAGE	1. IDEATION & INTEREST	2. LOCAL TESTING	3. TESTNET DEPLOY	4. MAINNET DEPLOY	5. SCALING DAPP USAGE
Action	Developer becomes interested in web3 and forms an idea of what to build	Developer installs libraries for web3 application building and local deployment	Developer connects to a testnet, gets test ETH from a faucet, and deploys their smart contracts	Developer connects to mainnet blockchain, places Ethereum in wallet and deploys to Mainnet	Developer maintains, improves and scales existing codebase for growth
Measurement	Alchemy Dapp Store, Alchemy Ventures,	Library downloads	sepoliafaucet.com, goerlifaucet.com, testnet contracts	Mainnet Contracts, EVM Contracts, L2 Bridging	Alchemy API Usage, Ecosystem Usage

Measuring builder momentum

These 3 indicators show us how devs are **building, deploying, and scaling**

Libraries

Developer tools to easily read/write to the blockchain

Smart contracts

Computer programs stored on a blockchain

Dapps

End-user ready applications with decentralized backends

1 Ideation and interest

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Positive developer sentiment and fundraising across fast-growing verticals point to continued interest in building web3 products.

alchemy Ventures

Startups focused on security and scaling

Q2 vertical segmentation for Alchemy Ventures focused on security, scaling, and infrastructure improvements.

INTMAX
ZettaBlock
FORDEFI
webacy
Cymbal
RISC ZERO
dora

Founders focused on

20% Security

30% Infrastructure

20% Transaction Infra

10% Layer 3 Technology

10% ZK Tech

10% Decentralized Idenity

Data driven dapps got a boost 🚀

2,000+

web3 solutions and counting
now on Alchemy Dapp Store:

+20%
since end of Q1

331
new dapps approved in Q2

Fastest growing categories

- Blockchain auditing
- Web3 data companies

+21% Q/Q

token terminal

SPACEANDTIME™

arbitrary execution

verichains

subsquid

Alchemy Q2 Developer Survey

- In Q2 2023, Alchemy surveyed web3 developers to gather insights into platform usage and market outlook.
- In total, 625 respondents provided feedback and insights on their interests and main blockers for growth.

Alchemy Q2 Developer Survey

“What topic in web3 are you most interested in?”

People were more interested in security and scalability than general tech trends like AI.

1. Smart contract security
2. Account abstraction
3. ZK-rollups
4. Smart contract development
5. AI

Alchemy Q2 Developer Survey

“What are the top obstacles to growth in the next year?”

People viewed funding as a bigger barrier to growth than regulation.

1. Funding
2. Awareness
3. Regulation
4. Resource Bandwidth

Web3 use cases in action

Fashion

Nike launched Our Force 1, an NFT experience that digitizes the option to buy a shoe as an NFT until it is redeemed for a physical shoe.

Social

Lens Protocol, the decentralized social media app, launched Momoka, its Optimistic L3 scaling solution that will process transactions at hyperscale.

Gaming

Google Play now allows developers to integrate blockchain-based experiences including NFTs and the use of tokenized assets within its apps and games.

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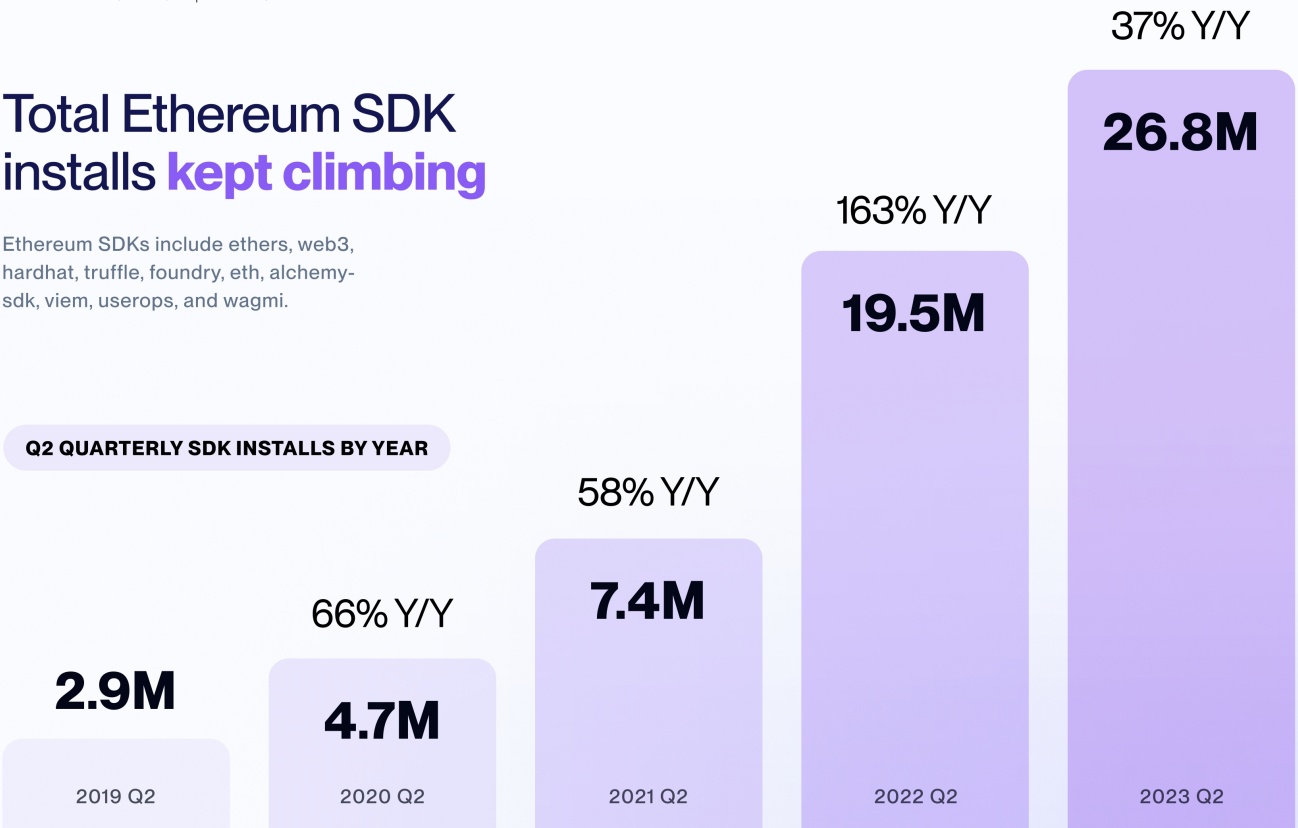
Before launching projects on testnets or mainnet blockchains, devs build on their personal computers (i.e. locally) to test and iterate before shipping.

Total Ethereum SDK installs kept climbing

Ethereum SDKs include ethers, web3, hardhat, truffle, foundry, eth, alchemy-sdk, viem, userops, and wagmi.

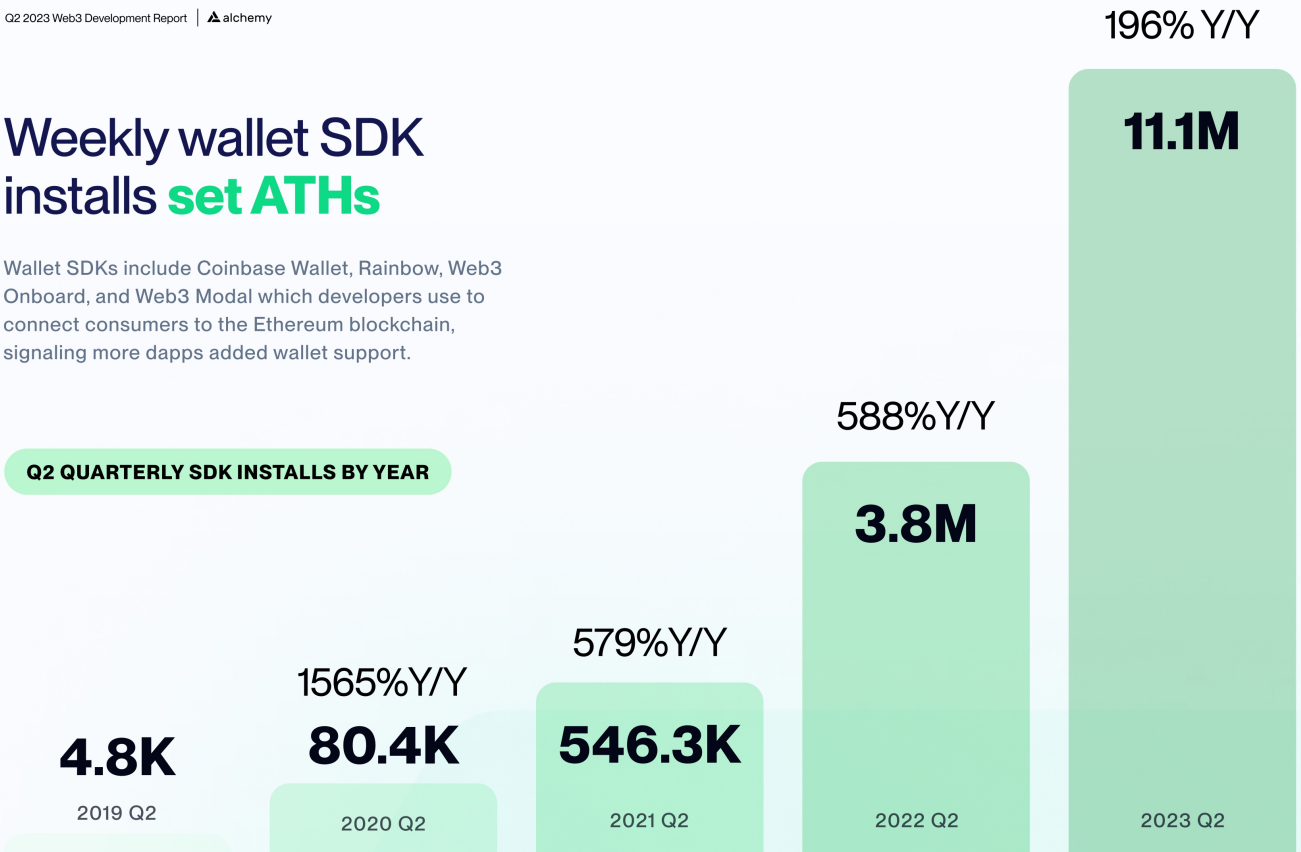
Q2 QUARTERLY SDK INSTALLS BY YEAR



Weekly wallet SDK installs set ATHs

Wallet SDKs include Coinbase Wallet, Rainbow, Web3 Onboard, and Web3 Modal which developers use to connect consumers to the Ethereum blockchain, signaling more dapps added wallet support.

Q2 QUARTERLY SDK INSTALLS BY YEAR



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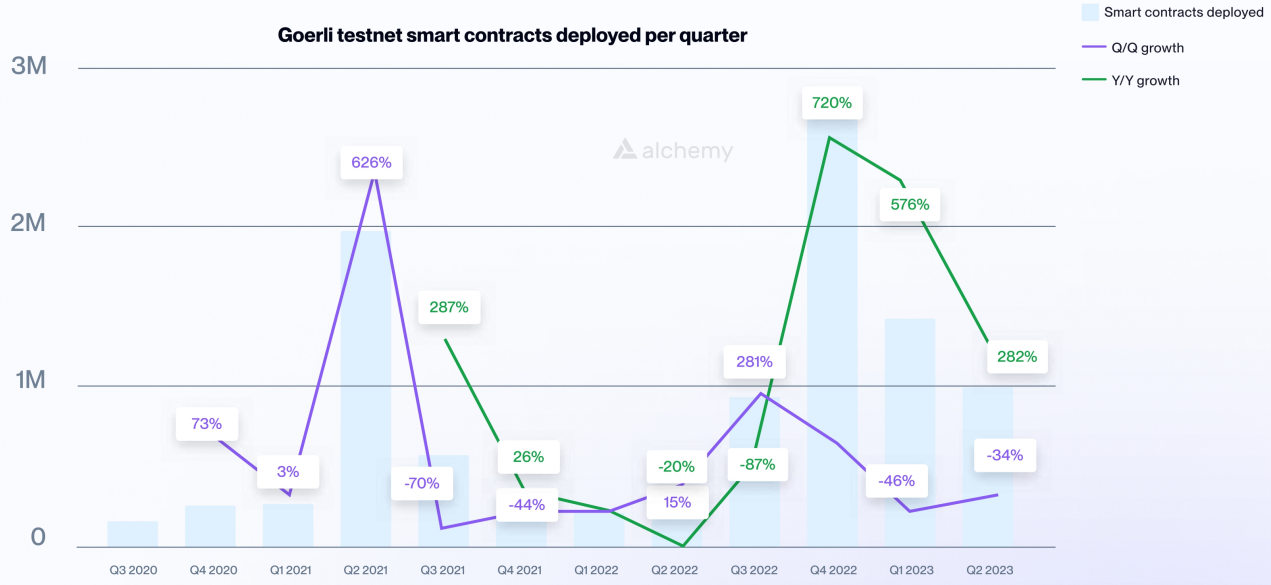
Developers deploy applications to testnets to refine their product on a blockchain that mirrors the mainnet environment without spending real ETH.

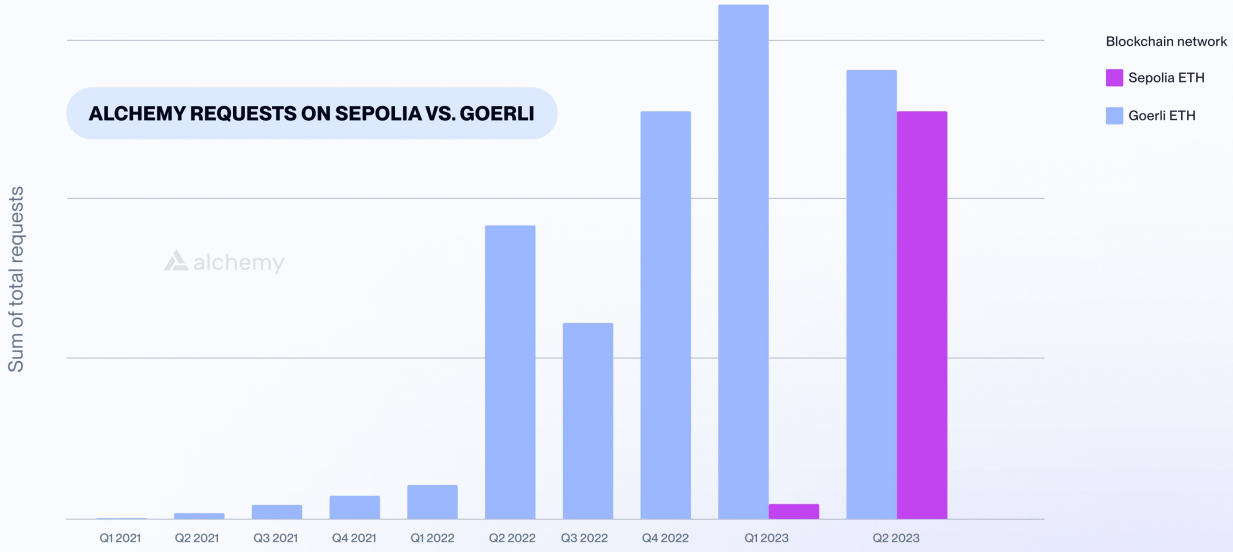
Goerli contracts dip as devs migrate to Sepolia

The emergence of Goerli ETH secondary markets in Q1 created a scarce supply, resulting in fewer Goerli contracts created in Q2.



Source: Dune

Sepolia flippening is approaching 🥇



Source: Alchemy

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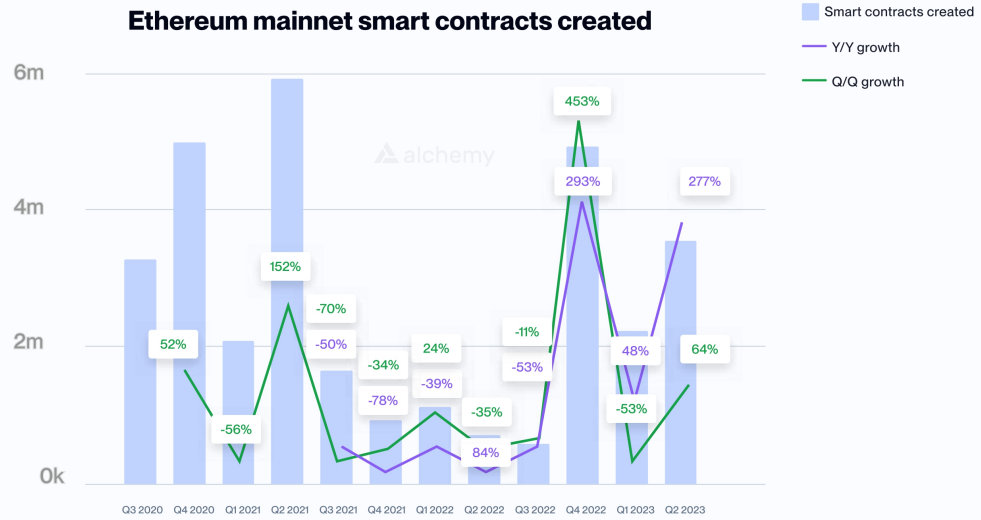
Developers deploy tested applications to the Ethereum Mainnet and Layer 2s when their product is ready to serve real users in a live environment.

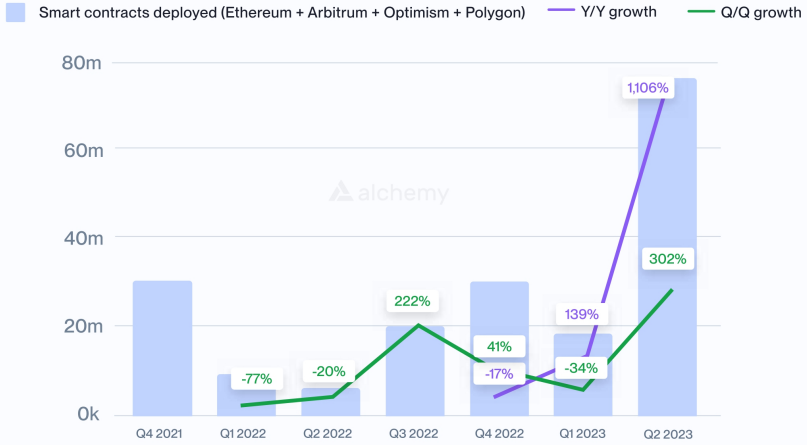
Ethereum smart contract deploys **grew Y/Y and Q/Q**



Source: Dune

EVM (Mainnet + Layer 2's) contracts **grew 1,106% Y/Y**

With gas remaining affordable on Polygon, whale contract creators deployed
+54M smart contracts created in Q2 alone.



Source: Dune

L2 bridging stayed steady Q/Q

Despite lower DEX and NFT trading volumes as a whole, bridging from Ethereum Mainnet to L2s remained steady Q/Q.

Source: Dune

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As web3 products attract real users and find product-market fit, developers need to scale their systems to support the increased consumer demand.

polygon

Polygon activity on Alchemy pops

Total requests

+33% Y/Y

Free requests

+70% Y/Y

Enhanced API requests

+535% Y/Y

Active teams

+74% Y/Y

Free teams

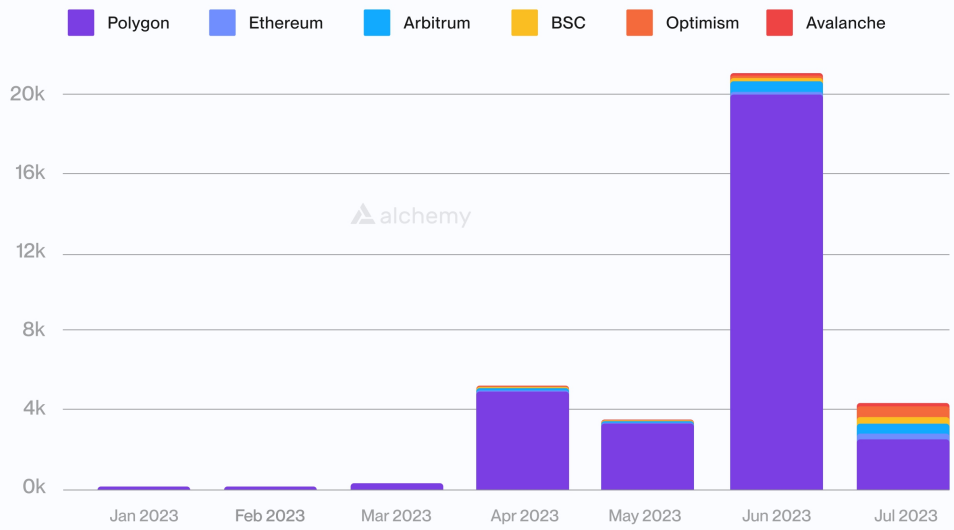
+73% Y/Y

Source: Alchemy

polygon

Account abstraction is gaining momentum on Polygon

541% increase in monthly active ERC-4337 Smart Contract Accounts



Source: Dune, 0xKofi

ARBITRUM

Arbinauts arrived en masse on Alchemy

Total requests

+295% Y/Y

Free requests

+485% Y/Y

Enhanced API requests

+7,589% Y/Y

Active teams

+213% Y/Y

Free teams

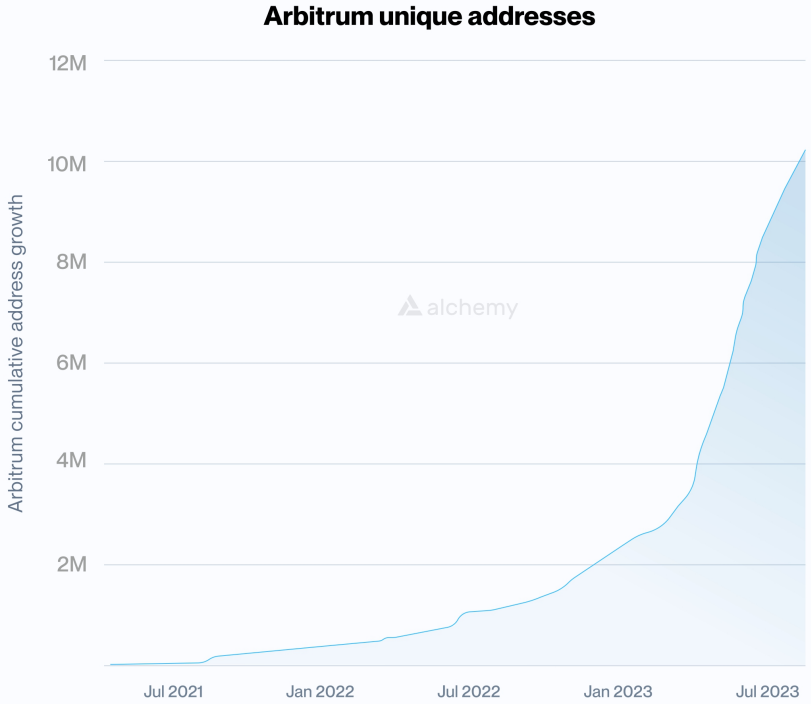
+216% Y/Y

Source: Alchemy

ARBITRUM

Arbitrum's Layer 2 boomed

Arbitrum One, an optimistic rollup, hit 9.5M unique addresses and more than doubled in Q2.



Source: Arbiscan.io

OPTIMISM

Optimism builders skyrocketed on Alchemy

Total requests

+110% Y/Y

Free requests

+637% Y/Y

Enhanced API requests

+4,983% Y/Y

Active teams

+205% Y/Y

Free teams

+212% Y/Y

Source: Alchemy

OPTIMISM

And big upgrades ensured a bright future ahead

The Bedrock upgrade enabled Optimism network to have lower fees, reduced deposit times, and improved node performance.

55%

gas savings per transaction

$0.15

average gas fee post-bedrock

$1.75M

saved in fees in first 30 days

Source: OP Mainnet

SOLANA

Solana devs on Alchemy soared to new heights

Total requests

+164%

Q2 2023 / Q3 2022

Free requests

+197%

Q2 2023 / Q3 2022

Active teams

+56%

Q2 2023 / Q3 2022

Free teams

+59%

Q2 2023 / Q3 2022

Source: Alchemy | Note: First full quarter of Solana Mainnet support was Q3 2022

ASTAR

Astar gets a jumpstart on Alchemy

Total requests

+7%

Q2 2023 / Q3 2022

Active teams

+175%

Q2 2023 / Q3 2022

Free teams

+114%

Q2 2023 / Q3 2022

Source: Alchemy | Note: First full quarter of Astar support was Q2 2023

ASTAR

And hit some major milestones

500K
Astar holders 🎉

20K
stakers 💰

+

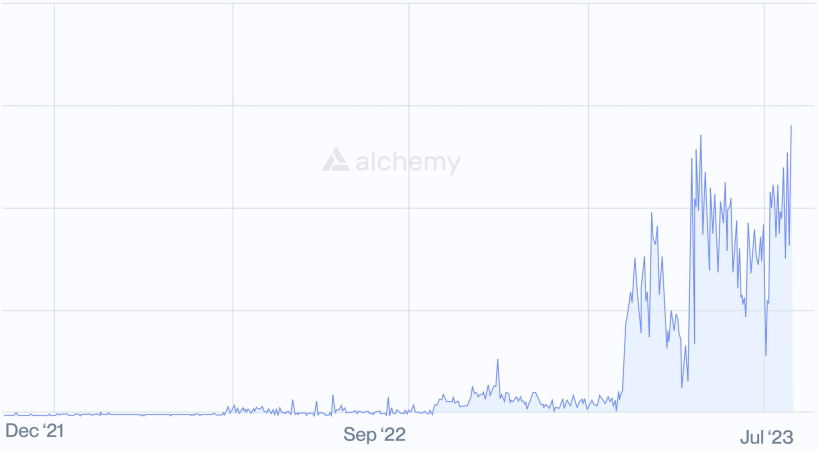
🚀 launched
Smart Contracts 2.0

Source: Astar

STARKNET

Starknet secured huge network gains

Starknet saw **10x growth in Transactions Per Second (TPS)** with their focus on boosting performance to new heights and delivering a seamless UX to attract developers.



Source: https://starkscan.co/stats

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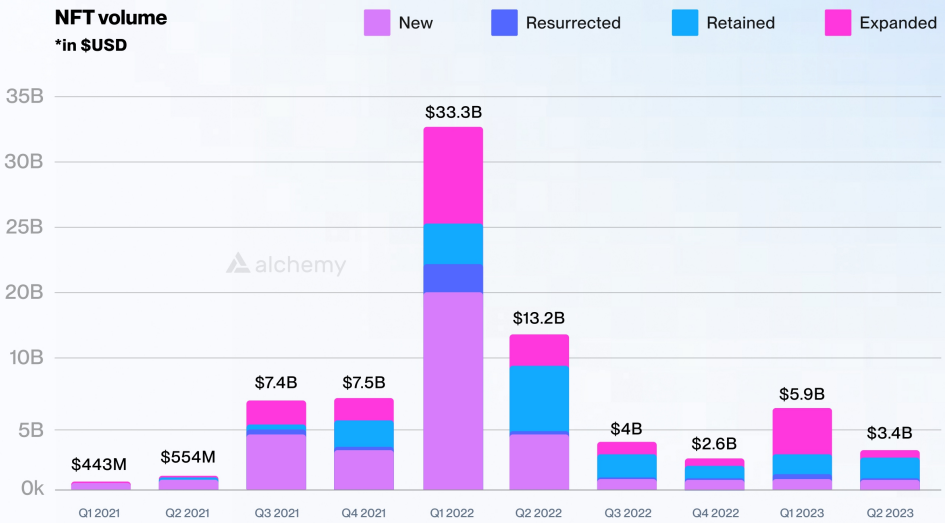
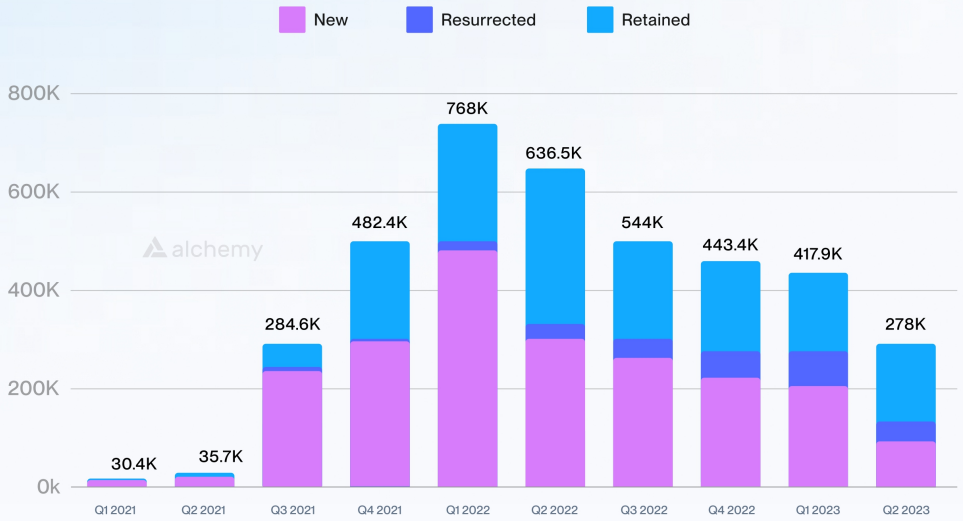
From dips in NFT trading activity to encouraging growth in DeFi users, here are a few other trends we found interesting in Q2.

NFT volume down -42% Q/Q



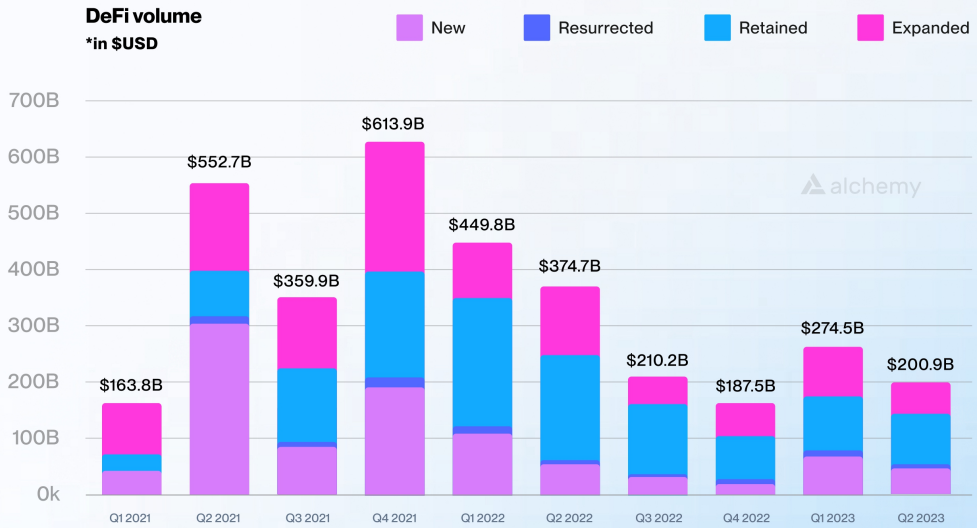
Source: Dune

NFT users down -33% Q/Q



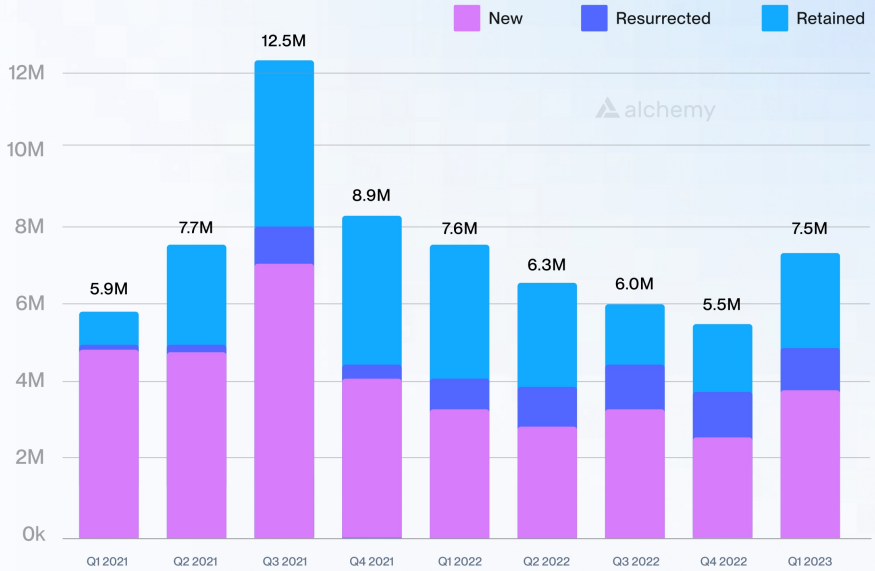
Source: Dune

DeFi volume down -27% Q/Q



Source: Dune

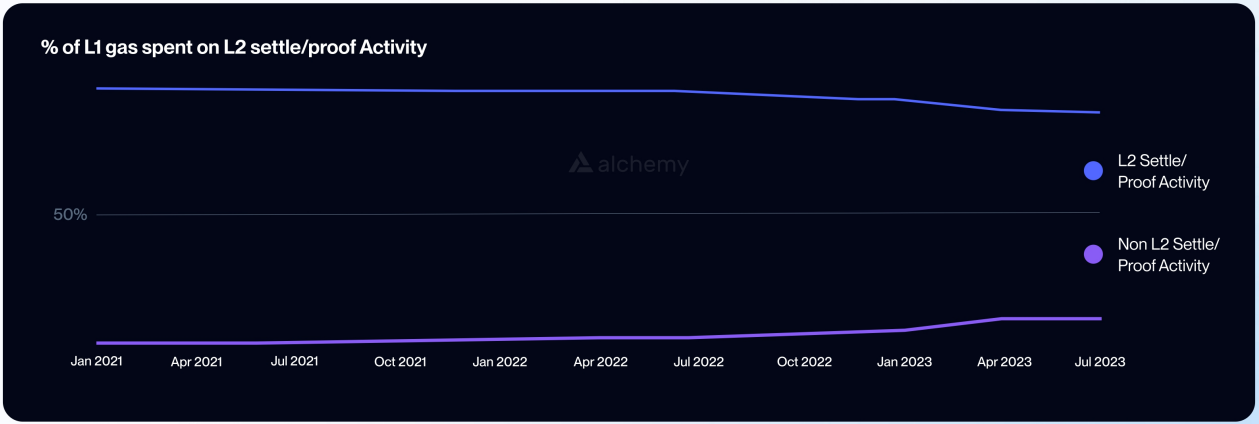
DeFi users up +35% Q/Q



Source: Dune

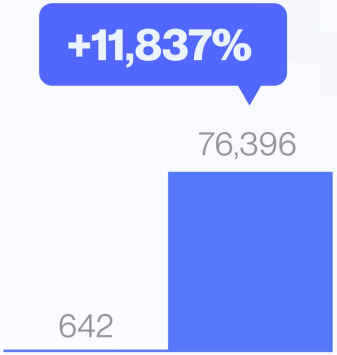
In Q2 2023, 9% of L1 gas usage was L2 related

L1 gas usage related to L2s grew 85% Q/Q and 383% Y/Y

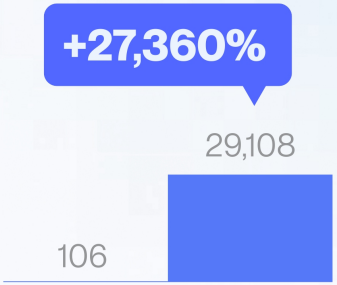


Source: Dune

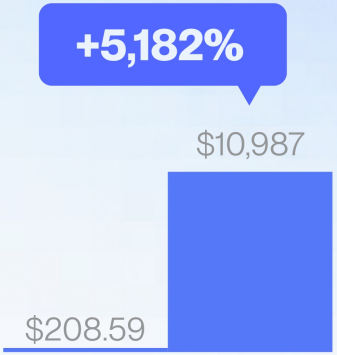
ERC-4337 takes off 🚀



Account Abstraction quarterly UserOps



Account Abstraction quarterly active users



Account Abstraction quarterly paymaster volume in USD

Source: Dune, 0xKofi

Methodology

Our aim

This report aims to provide an accurate and useful representation of web3 development. The following sources were used for data collection purposes: Alchemy's data, Dune (verified smart contracts), npmjs.com (SDK installations and metadata), and Coingecko (token prices).

Thanks to our sources

Smart Contracts, Bridging, and Programs: Dune. Special thanks to 0xK0fi.

Market and Prices: DappRadar, Defi Pulse, CoinMarketCap, Dune

SDKs: npmjs.com

For developer tools, resources and support, head to:

alchemy.com

For more web3 alpha and analysis, join our community:

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